

Introduction to Electronic Design Automation

Homework 2

(Problems 6-8)

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6 Two-level Logic Minimization

6. (a)

Use variable order $abcde$.

$$G = abce + abde + ab\bar{c}e + \bar{a}b\bar{c}e + \bar{a}\bar{b}\bar{c}\bar{e} + \bar{a}\bar{c}\bar{d}e + bc\bar{d}e + cd\bar{e}$$

$$D = \bar{a}b\bar{d}e + abcd\bar{e} + ac\bar{e}$$

Expanding them into minterms:

$$G = \{0, 1, 2, 4, 9, 11, 12, 13, 20, 25, 27, 28, 29, 31\}$$

$$D = \{9, 13, 20, 22, 28, 29, 30\}$$

Thus, we have the minterms of onset F as:

$$F = G - D = \{0, 1, 2, 4, 11, 12, 25, 27, 31\}$$

Therefore,

$$F \cup D = \{0, 1, 2, 4, 9, 11, 12, 13, 20, 22, 25, 27, 28, 29, 30, 31\}$$

$$\begin{aligned} R &= \{0, 1, \dots, 31\} - (F \cup D) \\ &= \{3, 5, 6, 7, 8, 10, 14, 15, 16, 17, 18, 19, 21, 23, 24, 26\} \end{aligned}$$

Writing them as product terms:

$$\begin{aligned} R &= \bar{a}\bar{b}\bar{c}de + \bar{a}\bar{b}c\bar{d}e + \bar{a}\bar{b}cd\bar{e} + \bar{a}\bar{b}cde + \bar{a}b\bar{c}\bar{d}\bar{e} + \bar{a}b\bar{c}d\bar{e} \\ &\quad + \bar{a}b\bar{c}d\bar{e} + \bar{a}b\bar{c}de + \bar{a}b\bar{c}\bar{d}\bar{e} + \bar{a}b\bar{c}d\bar{e} + \bar{a}b\bar{c}de + \bar{a}b\bar{c}d\bar{e} \\ &\quad + \bar{a}b\bar{c}de + \bar{a}b\bar{c}d\bar{e} + \bar{a}b\bar{c}de + \bar{a}b\bar{c}d\bar{e} \end{aligned}$$

6. (b)

First, we apply the Quine–McCluskey tabular method on $F \cup D$ to generate all prime implicants.

Round 1	Round 2	Round 3
$\bar{a}\bar{b}\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{a}\bar{b}\bar{c}\bar{d}$ $\bar{a}\bar{b}\bar{c}\bar{e}$ $\bar{a}\bar{b}\bar{d}\bar{e}$	
$\bar{a}\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{a}\bar{b}c\bar{d}\bar{e} \checkmark$ $\bar{a}b\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{a}\bar{c}\bar{d}\bar{e}$ $\bar{b}c\bar{d}\bar{e} \checkmark$ $\bar{a}c\bar{d}\bar{e} \checkmark$	
$\bar{a}\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{a}b\bar{c}\bar{d}\bar{e} \checkmark$ $a\bar{b}\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{a}\bar{b}\bar{c}e \checkmark$ $\bar{a}bde \checkmark$ $\bar{b}\bar{c}\bar{d}e \checkmark$ $\bar{a}bcd \checkmark$ $\bar{b}c\bar{d}\bar{e} \checkmark$ $\bar{a}b\bar{c}e \checkmark$ $ac\bar{d}\bar{e} \checkmark$	$c\bar{d}\bar{e}$ $b\bar{d}e$ $b\bar{c}e$ bcd $ac\bar{e}$
$\bar{a}\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{a}b\bar{c}d\bar{e} \checkmark$ $a\bar{b}\bar{c}d\bar{e} \checkmark$ $a\bar{b}c\bar{d}\bar{e} \checkmark$ $ab\bar{c}\bar{d}\bar{e} \checkmark$	$\bar{b}\bar{c}d\bar{e} \checkmark$ $\bar{b}c\bar{d}\bar{e} \checkmark$ $ac\bar{e} \checkmark$ $ab\bar{c}e \checkmark$ $ab\bar{d}e \checkmark$ $abcd \checkmark$ $abc\bar{e} \checkmark$	abe abc
$ab\bar{c}d\bar{e} \checkmark$ $abc\bar{d}\bar{e} \checkmark$ $abcd\bar{e} \checkmark$	$abde \checkmark$ $abce \checkmark$ $abcd \checkmark$	
$abcde \checkmark$		

Table 1: Quine–McCluskey tabular method for generating prime implicants

Thus, the prime implicants are

$$\{\bar{a}\bar{b}\bar{c}\bar{d}, \bar{a}\bar{b}\bar{c}\bar{e}, \bar{a}\bar{b}\bar{d}\bar{e}, \bar{a}\bar{c}\bar{d}\bar{e}, c\bar{d}\bar{e}, b\bar{d}e, \bar{b}\bar{c}e, bcd, ac\bar{e}, abe, abc\}.$$

Next, we use the Column covering of Boolean matrix to find the essential prime implicants. The Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{b}\bar{c}\bar{e}$	$\bar{a}\bar{b}\bar{d}\bar{e}$	$\bar{a}\bar{c}\bar{d}\bar{e}$	$c\bar{d}\bar{e}$	$b\bar{d}e$	$\bar{b}\bar{c}e$	bcd	$ac\bar{e}$	abe	abc
$m_0 = \bar{a}\bar{b}\bar{c}\bar{d}\bar{e}$	1	1	1	0	0	0	0	0	0	0	0
$m_1 = \bar{a}\bar{b}\bar{c}d\bar{e}$	1	0	0	1	0	0	0	0	0	0	0
$m_2 = \bar{a}\bar{b}c\bar{d}\bar{e}$	0	1	0	0	0	0	0	0	0	0	0
$m_4 = \bar{a}\bar{b}cd\bar{e}$	0	0	1	0	1	0	0	0	0	0	0
$m_{11} = \bar{a}b\bar{c}d\bar{e}$	0	0	0	0	0	0	1	0	0	0	0
$m_{12} = \bar{a}bcd\bar{e}$	0	0	0	0	1	0	0	1	0	0	0
$m_{25} = ab\bar{c}\bar{d}\bar{e}$	0	0	0	0	0	1	1	0	0	1	0
$m_{27} = ab\bar{c}d\bar{e}$	0	0	0	0	0	0	1	0	0	1	0
$m_{31} = abcde$	0	0	0	0	0	0	0	0	0	1	1

Table 2: Boolean matrix

6. (c)

Rows m_2 and m_{11} each have only one 1, so $\bar{a}\bar{b}\bar{c}\bar{e}$ and $b\bar{c}e$ are essential prime implicants. So we remove the rows m_2, m_{11} and the columns of $\bar{a}\bar{b}\bar{c}\bar{e}$ and $b\bar{c}e$. We also remove the rows that have 1 in the removed columns, which are m_0, m_{25}, m_{27} . After removing these rows, the columns $a\bar{c}\bar{e}$ and $b\bar{d}e$ contain only zeros, so we remove them as well. The resulting Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{b}\bar{d}\bar{e}$	$\bar{a}\bar{c}\bar{d}e$	$c\bar{d}\bar{e}$	$b\bar{c}\bar{d}$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	0	1	0	0	0	0
$m_4 = \bar{a}\bar{b}\bar{c}\bar{d}\bar{e}$	0	1	0	1	0	0	0
$m_{12} = \bar{a}b\bar{c}\bar{d}\bar{e}$	0	0	0	1	1	0	0
$m_{31} = abcde$	0	0	0	0	0	1	1

Table 3: Reduced Boolean matrix

6. (d)

Column $c\bar{d}\bar{e}$ covers rows m_4 and m_{12} . It dominates both columns $\bar{a}\bar{b}\bar{d}\bar{e}$ and $b\bar{c}\bar{d}$, so we remove these two dominated columns. The resulting Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{c}\bar{d}e$	$c\bar{d}\bar{e}$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	1	0	0	0
$m_4 = \bar{a}\bar{b}\bar{c}\bar{d}\bar{e}$	0	0	1	0	0
$m_{12} = \bar{a}b\bar{c}\bar{d}\bar{e}$	0	0	1	0	0
$m_{31} = abcde$	0	0	0	1	1

Table 4: Reduced Boolean matrix

Since rows m_4 and m_{12} have only one 1, the prime implicant $c\bar{d}\bar{e}$ is an induced essential prime implicant. Therefore, we select it and remove rows m_4 and m_{12} .

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{c}\bar{d}e$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	1	0	0
$m_{31} = abcde$	0	0	1	1

Table 5: Reduced Boolean matrix

6. (e)

After the reductions in (d), there is no cyclic core to solve, since the remaining matrix is decomposed into two independent single-row choices:

$$m_1 : \{\bar{a}\bar{b}\bar{c}\bar{d}, \bar{a}\bar{c}\bar{d}e\}$$

$$m_{31} : \{abe, abc\}$$

Thus, the independent set heuristic is not applicable. We only need to choose one prime implicant from each set. For example, we choose

$$\bar{a}\bar{b}\bar{c}\bar{d} \quad \text{and} \quad abe.$$

Therefore, the selected prime implicants are

$$\bar{a}\bar{b}\bar{c}\bar{e}, \quad b\bar{c}e, \quad c\bar{d}\bar{e}, \quad \bar{a}\bar{b}\bar{c}\bar{d}, \quad abe.$$

6. (f)

There are four minimum covers with 5 prime implicants:

$$G_1^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{b}\bar{c}\bar{d} + abe.$$

$$G_2^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{c}\bar{d}e + abe.$$

$$G_3^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{b}\bar{c}\bar{d} + abc.$$

$$G_4^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{c}\bar{d}e + abc.$$

7 Static Timing Analysis**7. (a)**

In this part, we construct the circuit as a DAG, where each node $v \in V$ represents a gate and each edge $e \in E \subseteq V \times V$ represents a connection between gates. Let $R(v)$ denote the required time at the input of node v , $A(v)$ denote the arrival time at the output of node v , and $d_{u,v}$ denotes the delay from node u 's output to node v 's output. Also, we define the slack of node v as

$$S(v) = R(v) - A(v).$$

Suppose there is a node v such that

$$S(v) = R(v) - A(v) \leq c$$

We want to show that there exists a path satisfying every node on the path has slack at most c from some primary input to v and from v to some primary output.

1. From some primary input to v

First, if v is a primary input, then we are done. Otherwise, since

$$A(v) = \max_{u:(u,v) \in E} (A(u) + d_{u,v})$$

there exist a node u such that there is an edge from u to v and

$$A(v) = A(u) + d_{u,v}.$$

And since

$$R(u) = \min_{w:(u,w) \in E} (R(w) - d_{u,w}) \leq R(v) - d_{u,v},$$

we have

$$S(u) = R(u) - A(u) \leq R(v) - d_{u,v} - A(u) = R(v) - A(v) = S(v) \leq c.$$

Thus, we can apply the same argument on u and repeat this process until we reach a primary input.

2. From v to some primary output

First, if v is a primary output, then we are done. Otherwise, since

$$R(v) = \min_{w:(v,w) \in E} (R(w) - d_{v,w})$$

there exist a node w such that there is an edge from v to w and

$$R(v) = R(w) - d_{v,w}.$$

And since

$$A(w) = \max_{x:(x,w) \in E} (A(x) + d_{x,w}) \geq A(v) + d_{v,w},$$

we have

$$S(w) = R(w) - A(w) \leq R(v) + d_{v,w} - A(v) - d_{v,w} = R(v) - A(v) = S(v) \leq c.$$

Thus, we can apply the same argument on w and repeat this process until we reach a primary output.

Hence, there exists a path from some primary input to v and from v to some primary output such that the slack of each node on these paths is at most c .

7. (b)

The equivalence DAG of the given circuit is as Figure 1 below. The delay of each gate is labeled on the corresponding node. Arrival time, required time and slack of each node are also labeled on the corresponding node in the format of (a, r, s) , where a is the arrival time, r is the required time and s is the slack. The critical path is highlighted in red.

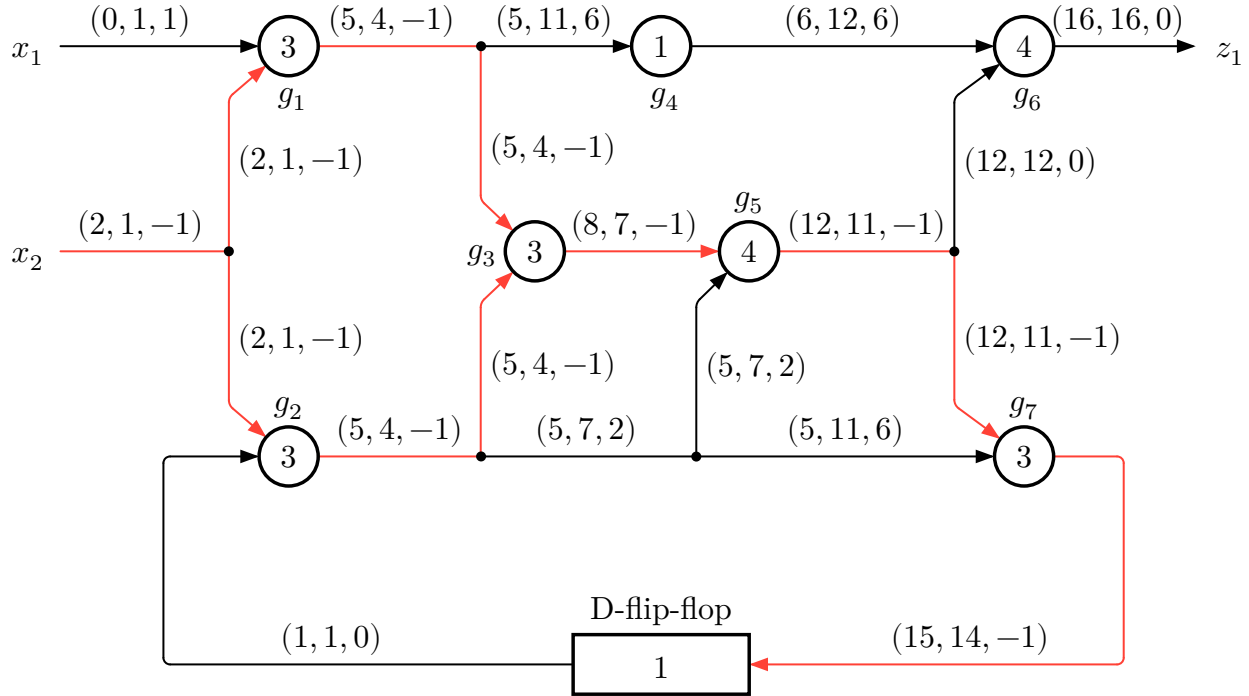


Figure 1: static timing analysis result

8 Technology Mapping

In this problem, I wrote two versions of the program. The first one is the basic DAGON mapping algorithm. The second one is an improved version, where I insert two inverters on every original edge before running DAGON. This gives the mapper more chance to use `INV` and `buf`.

The node numbering I used in my input file is shown in Figure 2.

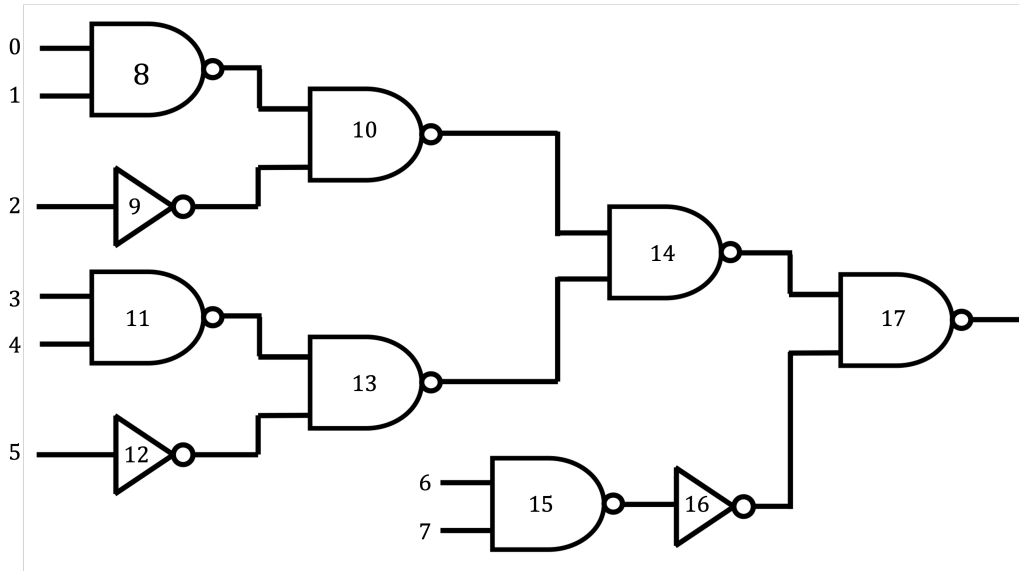


Figure 2: Node numbering used in `input.txt`

First, for the basic version without improvement, the result is:

Node	Minimum cost	Selected cell
8	2	nand2
9	1	inv
10	5	nand2
11	2	nand2
12	1	inv
13	5	nand2
14	12	nand2
15	2	nand2
16	3	inv
17	15	nand3

Table 6: Optimal cost and selected cell of each subject node

Thus, the final cover without improvement is obtained by tracing back from output node 17:

$\{8 : \text{nand2}, 9 : \text{inv}, 10 : \text{nand2}, 11 : \text{nand2}, 12 : \text{inv}, 13 : \text{nand2}, 14 : \text{nand2}, 17 : \text{nand3}\}$.

The `nand3` at node 17 covers nodes 15, 16, and 17 together. Therefore, the minimum total area is

$$2 + 1 + 2 + 2 + 1 + 2 + 2 + 3 = 15.$$

Next, for the improved version, the output on original nodes is:

Node	Minimum cost	Selected cell
8	2	nand2
9	1	inv
10	5	nand2
11	2	nand2
12	1	inv
13	5	nand2
14	10	nand2
15	2	nand2
16	3	inv
17	12	oai21

Table 7: Improved result on original subject nodes

The minimum total area becomes 12. This is smaller than 15 because the inserted inverter pairs create extra matching choices. In particular, node 17 can be matched by `oai21` in the improved graph, so the final cost is reduced. However, if we only apply the original DAGON algorithm directly on the given subject graph, the answer is still 15.

The c++ code of both versions and the input/output files are included in the appendix.

Appendix: `tagon.cpp`

cpp

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 #define ll long long
4 #define pb push_back
5 #define endl '\n'
6 #define AI(x) begin(x),end(x)
7
8
9 const int INF = 1e9;
10
11 enum class GateType {
12     PI,
13     INV,
14     NAND,
15     NOR,
16 };
17
18 string toUpper(string s) {
19     for (char &c : s) c = toupper(c);
20     return s;
21 }
22
23 GateType s2gate_type(string s) {
24     s = toUpper(s);
25
26     if (s == "PI" || s == "IN") return GateType::PI;
27     if (s == "INV") return GateType::INV;
28     if (s == "NAND") return GateType::NAND;
29     if (s == "NOR") return GateType::NOR;
30
31     cerr << "Unknown gate type: " << s << endl;
32     exit(1);
33 }
34
35 bool isCommutative(GateType t) {
36     return t == GateType::NAND || t == GateType::NOR;
37 }
38
39 struct node {
40     int id;
41     GateType type;
42     vector<int> fanin;

```

```

43     vector<int> fanout;
44
45     node() {}
46     node(int i, GateType g) : id(i), type(g) {}
47 };
48
49 vector<int> topological_sort(vector<node>& g) {
50     int n = g.size();
51     vector<int> indeg(n), order;
52     queue<int> q;
53
54     for (int i = 0; i < n; i++) {
55         indeg[i] = g[i].fanin.size();
56         if (indeg[i] == 0) q.push(i);
57     }
58
59     while (!q.empty()) {
60         int u = q.front();
61         q.pop();
62
63         order.pb(u);
64
65         for (int v : g[u].fanout) {
66             indeg[v]--;
67             if (indeg[v] == 0) q.push(v);
68         }
69     }
70
71     return order;
72 }
73
74 // 回傳 pattern subtree match 到 subject subtree 後, boundary dp cost 總和
75 // 若不能 match, 回傳 INF
76 int matchCost(int sid, int pid,
77               vector<node>& sg,
78               vector<node>& pg,
79               vector<int>& dp) {
80
81     node& s = sg[sid];
82     node& p = pg[pid];
83
84     if (p.type == GateType::PI) {
85         return dp[sid];
86     }
87
88     if (s.type != p.type) return INF;
89     if (s.fanin.size() != p.fanin.size()) return INF;
90
91     int n = s.fanin.size();
92
93     vector<int> idx(n);
94     iota(idx.begin(), idx.end(), 0);
95
96     int best = INF;
97
98     do {
99         int total = 0;
100         bool ok = true;
101
102         for (int i = 0; i < n; i++) {
103             int s_child = s.fanin[idx[i]];
104             int p_child = p.fanin[i];
105
106             int sub = matchCost(s_child, p_child, sg, pg, dp);
107
108             if (sub >= INF) {
109                 ok = false;
110                 break;
111             }
112
113             total += sub;
114         }
115
116         if (ok) best = min(best, total);
117     } while (isCommutative(s.type) &&
118             next_permutation(idx.begin(), idx.end()));
119
120     return best;
121 }
122
123
124 int32_t main() {
125     ios::sync_with_stdio(0); cin.tie(0); cout.tie(0);
126     int ns, ms;
127     cin >> ns >> ms;
128
129     vector<node> sg(ns);
130
131     for (int i = 0; i < ns; i++) {
132         string t;

```

```

133     cin >> t;
134     sg[i] = node(i, s2gate_type(t));
135 }
136
137 for (int i = 0; i < ns; i++) {
138     int a, b;
139     cin >> a >> b;
140
141     sg[a].fanout.pb(b);
142     sg[b].fanin.pb(a);
143 }
144
145 vector<int> subject_order = topological_sort(sg);
146
147 int p;
148 cin >> p;
149
150 vector<vector<node>> patterns(p);
151 vector<int> cost(p);
152 vector<string> name(p);
153 vector<int> root(p);
154
155 for (int i = 0; i < p; i++) {
156     cin >> name[i] >> cost[i];
157
158     int n, m, r;
159     cin >> n >> m >> r;
160
161     root[i] = r;
162     patterns[i].resize(n);
163
164     for (int j = 0; j < n; j++) {
165         string t;
166         cin >> t;
167         patterns[i][j] = node(j, s2gate_type(t));
168     }
169
170     for (int j = 0; j < m; j++) {
171         int a, b;
172         cin >> a >> b;
173
174         patterns[i][a].fanout.pb(b);
175         patterns[i][b].fanin.pb(a);
176     }
177 }
178
179 vector<int> dp(ns, INF);
180 vector<int> choose(ns, -1);
181
182 for (int v : subject_order) {
183     if (sg[v].type == GateType::PI) {
184         dp[v] = 0;
185         continue;
186     }
187
188     for (int i = 0; i < p; i++) {
189         int boundary_cost = matchCost(v, root[i], sg, patterns[i], dp);
190
191         if (boundary_cost >= INF) continue;
192
193         int total_cost = cost[i] + boundary_cost;
194
195         if (total_cost < dp[v]) {
196             dp[v] = total_cost;
197             choose[v] = i;
198         }
199     }
200 }
201
202 // 預設所有 fanout empty 的 node 都是 output
203 int ans = 0;
204
205 for (int i = 0; i < ns; i++) {
206     if (sg[i].fanout.empty()) {
207         ans += dp[i];
208     }
209 }
210
211 cout << "Minimum total cost = " << ans << endl;
212
213 for (int i = 0; i < ns; i++) {
214     cout << "node " << i << ": ";
215
216     if (sg[i].fanin.empty()) {
217         cout << "PI, cost = 0" << endl;
218     } else {
219         cout << "cost = " << dp[i];
220
221         if (choose[i] != -1) {
222             cout << ", cell = " << name[choose[i]];

```

```

223         } else {
224             cout << ", cell = NONE";
225         }
226
227         cout << endl;
228     }
229 }
230
231 return 0;
232 }
233

```

Appendix: dagon_improve.cpp

cpp

```

1 // dagon_improve.cpp
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 #define ll long long
6 #define pb push_back
7 #define endl '\n'
8
9 const ll INF = (ll)4e18;
10
11 enum class GateType {
12     PI,
13     INV,
14     NAND,
15     NOR,
16 };
17
18 string toUpper(string s) {
19     for (char &c : s) c = toupper(c);
20     return s;
21 }
22
23 GateType s2gate_type(string s) {
24     s = toUpper(s);
25
26     if (s == "PI" || s == "IN") return GateType::PI;
27     if (s == "INV") return GateType::INV;
28     if (s == "NAND") return GateType::NAND;
29     if (s == "NOR") return GateType::NOR;
30
31     cerr << "Unknown gate type: " << s << endl;
32     exit(1);
33 }
34
35 string gate2str(GateType t) {
36     if (t == GateType::PI) return "PI";
37     if (t == GateType::INV) return "INV";
38     if (t == GateType::NAND) return "NAND";
39     if (t == GateType::NOR) return "NOR";
40     return "?";
41 }
42
43 bool isCommutative(GateType t) {
44     return t == GateType::NAND || t == GateType::NOR;
45 }
46
47 struct Node {
48     int id;
49     GateType type;
50     vector<int> fanin;
51     vector<int> fanout;
52
53     Node() {}
54     Node(int i, GateType g) : id(i), type(g) {}
55 };
56
57 struct Pattern {
58     string name;
59     ll cost;
60     int root;
61     vector<Node> g;
62 };
63
64 void addEdge(vector<Node> &g, int u, int v) {
65     g[u].fanout.pb(v);
66     g[v].fanin.pb(u);
67 }
68
69 vector<int> topological_sort(vector<Node> &g) {
70     int n = (int)g.size();
71     vector<int> indeg(n), order;
72     queue<int> q;
73
74     for (int i = 0; i < n; i++) {

```

```

75     indeg[i] = (int)g[i].fanin.size();
76     if (indeg[i] == 0) q.push(i);
77 }
78
79 while (!q.empty()) {
80     int u = q.front();
81     q.pop();
82
83     order.pb(u);
84
85     for (int v : g[u].fanout) {
86         if (--indeg[v] == 0) q.push(v);
87     }
88 }
89
90 if ((int)order.size() != n) {
91     cerr << "Error: graph is not a DAG." << endl;
92     exit(1);
93 }
94
95 return order;
96 }
97
98 // 對 subject graph 每條原始 edge u -> v 插入兩個 INV :
99 // u -> inv1 -> inv2 -> v
100 void insertInverterPairs(vector<Node> &sg,
101                          const vector<pair<int,int>> &edges,
102                          vector<char> &isInserted) {
103     for (auto &x : sg) {
104         x.fanin.clear();
105         x.fanout.clear();
106     }
107
108     for (auto [u, v] : edges) {
109         int inv1 = (int)sg.size();
110         sg.pb(Node(inv1, GateType::INV));
111         isInserted.pb(true);
112
113         int inv2 = (int)sg.size();
114         sg.pb(Node(inv2, GateType::INV));
115         isInserted.pb(true);
116
117         addEdge(sg, u, inv1);
118         addEdge(sg, inv1, inv2);
119         addEdge(sg, inv2, v);
120     }
121 }
122
123 // 回傳 pattern subtree match 到 subject subtree 後的 boundary dp cost 總和。
124 // 若不能 match, 回傳 INF.
125 ll matchCost(int sid, int pid,
126              vector<Node> &sg,
127              vector<Node> &pg,
128              vector<ll> &dp,
129              vector<char> &isInserted) {
130
131     Node &s = sg[sid];
132     Node &pnode = pg[pid];
133
134     // pattern 的 PI 是 boundary, 可以停在任何 subject node
135     if (pnode.type == GateType::PI) {
136         return dp[sid];
137     }
138
139     ll bestSkip = INF;
140
141     // 嘗試跳過 virtual INV pair:
142     // u -> inv1 -> inv2
143     // sid = inv2 時, 可以直接嘗試 match u
144     if (isInserted[sid] &&
145         s.type == GateType::INV &&
146         s.fanin.size() == 1) {
147
148         int inv1 = s.fanin[0];
149
150         if (isInserted[inv1] &&
151             sg[inv1].type == GateType::INV &&
152             sg[inv1].fanin.size() == 1) {
153
154             int u = sg[inv1].fanin[0];
155
156             bestSkip = matchCost(u, pid, sg, pg, dp, isInserted);
157         }
158     }
159
160     if (s.type != pnode.type) return bestSkip;
161     if (s.fanin.size() != pnode.fanin.size()) return bestSkip;
162
163     int n = (int)s.fanin.size();
164

```

```

165 vector<int> idx(n);
166 iota(idx.begin(), idx.end(), 0);
167
168 ll best = INF;
169
170 do {
171     ll total = 0;
172     bool ok = true;
173
174     for (int i = 0; i < n; i++) {
175         int s_child = s.fanin[idx[i]];
176         int p_child = pnode.fanin[i];
177
178         ll sub = matchCost(s_child, p_child, sg, pg, dp, isInserted);
179
180         if (sub >= INF) {
181             ok = false;
182             break;
183         }
184
185         total += sub;
186
187         if (total >= INF) {
188             ok = false;
189             break;
190         }
191     }
192
193     if (ok) best = min(best, total);
194 } while (isCommutative(s.type) &&
195         next_permutation(idx.begin(), idx.end()));
196
197 return min(best, bestSkip);
198 }
199
200 int32_t main() {
201     ios::sync_with_stdio(0); cin.tie(0); cout.tie(0);
202     int ns, ms;
203     cin >> ns >> ms;
204
205     vector<Node> sg(ns);
206
207     for (int i = 0; i < ns; i++) {
208         string t;
209         cin >> t;
210         sg[i] = Node(i, s2gate_type(t));
211     }
212
213     vector<pair<int,int>> original_edges;
214
215     for (int i = 0; i < ms; i++) {
216         int a, b;
217         cin >> a >> b;
218         original_edges.pb({a, b});
219         addEdge(sg, a, b);
220     }
221
222     // 記錄原始 output, 因為後面會清除並重建 edge
223     vector<int> original_outputs;
224     for (int i = 0; i < ns; i++) {
225         if (sg[i].fanout.empty()) original_outputs.pb(i);
226     }
227
228     // improve DAGON: 每條 wire 插入兩個 INV
229     vector<char> isInserted(ns, false);
230     insertInverterPairs(sg, original_edges, isInserted);
231
232     int p;
233     cin >> p;
234
235     vector<Pattern> patterns;
236
237     for (int i = 0; i < p; i++) {
238         Pattern pat;
239         cin >> pat.name >> pat.cost;
240
241         int n, m, r;
242         cin >> n >> m >> r;
243
244         pat.root = r;
245         pat.g.resize(n);
246
247         for (int j = 0; j < n; j++) {
248             string t;
249             cin >> t;
250             pat.g[j] = Node(j, s2gate_type(t));
251         }
252
253         for (int j = 0; j < m; j++) {

```

```

255     int a, b;
256     cin >> a >> b;
257     addEdge(pat.g, a, b);
258 }
259
260 patterns.pb(pat);
261 }
262
263 vector<int> order = topological_sort(sg);
264
265 int nsg = (int)sg.size();
266 vector<ll> dp(nsg, INF);
267 vector<int> choose(nsg, -1);
268
269 for (int v : order) {
270     if (sg[v].type == GateType::PI) {
271         dp[v] = 0;
272         continue;
273     }
274
275     for (int i = 0; i < (int)patterns.size(); i++) {
276         ll boundary_cost = matchCost(v, patterns[i].root, sg, patterns[i].g, dp, isInserted);
277
278         if (boundary_cost >= INF) continue;
279
280         ll total_cost = patterns[i].cost + boundary_cost;
281
282         if (total_cost < dp[v]) {
283             dp[v] = total_cost;
284             choose[v] = i;
285         }
286     }
287 }
288
289 ll ans = 0;
290 for (int out : original_outputs) {
291     ans += dp[out];
292 }
293
294 cout << "Minimum total cost = " << ans << endl;
295
296 cout << endl;
297 cout << "[Original nodes]" << endl;
298 for (int i = 0; i < ns; i++) {
299     cout << "node " << i << " (" << gate2str(sg[i].type) << "): ";
300
301     if (sg[i].type == GateType::PI) {
302         cout << "PI, cost = 0";
303     } else {
304         cout << "cost = " << dp[i];
305
306         if (choose[i] != -1) {
307             cout << ", cell = " << patterns[choose[i]].name;
308         } else {
309             cout << ", cell = NONE";
310         }
311     }
312
313     bool is_output = false;
314     for (int out : original_outputs) {
315         if (out == i) is_output = true;
316     }
317     if (is_output) cout << " [OUTPUT]";
318
319     cout << endl;
320 }
321
322 cout << endl;
323 cout << "[Inserted inverter-pair nodes]" << endl;
324
325 for (int i = ns; i < nsg; i++) {
326     cout << "node " << i << " (" << gate2str(sg[i].type) << "): ";
327     cout << "cost = " << dp[i];
328
329     if (choose[i] != -1) {
330         cout << ", cell = " << patterns[choose[i]].name;
331     } else {
332         cout << ", cell = NONE";
333     }
334
335     cout << endl;
336 }
337
338 return 0;
339 }
340

```

Appendix: input.txt

txt

```

1 18 17
2 PI PI PI PI PI PI PI PI NAND INV NAND INV NAND NAND INV NAND
3 0 8
4 1 8
5 2 9
6 8 10
7 9 10
8 3 11
9 4 11
10 5 12
11 11 13
12 12 13
13 10 14
14 13 14
15 6 15
16 7 15
17 15 16
18 14 17
19 16 17
20 10
21 inv 1
22 2 1 1
23 PI INV
24 0 1
25 buf 0
26 3 2 2
27 PI INV INV
28 0 1
29 1 2
30 nand2 2
31 3 2 2
32 PI PI NAND
33 0 2
34 1 2
35 and2 3
36 4 3 3
37 PI PI NAND INV
38 0 2
39 1 2
40 2 3
41 nand3 3
42 6 5 5
43 PI PI PI NAND INV NAND
44 0 3
45 1 3
46 3 4
47 4 5
48 2 5
49 or2 4
50 5 4 4
51 PI PI INV INV NAND
52 0 2
53 1 3
54 2 4
55 3 4
56 aoi21 3
57 7 6 6
58 PI PI PI NAND INV NAND INV
59 0 3
60 1 3
61 2 4
62 3 5
63 4 5
64 5 6
65 oai21 3
66 7 6 6
67 PI PI PI INV INV NAND NAND
68 0 3
69 1 4
70 3 5
71 4 5
72 5 6
73 2 6
74 aoi22 4
75 8 7 7
76 PI PI PI PI NAND NAND NAND INV
77 0 4
78 1 4
79 2 5
80 3 5
81 4 6
82 5 6
83 6 7
84 nand4 4
85 9 8 8
86 PI PI PI PI NAND INV NAND INV NAND
87 0 4
88 1 4
89 4 5
90 5 6
91 2 6
92 6 7
93 7 8
94 3 8

```

Appendix: output.txt

txt

```

1 Minimum total cost = 15
2 node 0: PI, cost = 0

```

```

3 node 1: PI, cost = 0
4 node 2: PI, cost = 0
5 node 3: PI, cost = 0
6 node 4: PI, cost = 0
7 node 5: PI, cost = 0
8 node 6: PI, cost = 0
9 node 7: PI, cost = 0
10 node 8: cost = 2, cell = nand2
11 node 9: cost = 1, cell = inv
12 node 10: cost = 5, cell = nand2
13 node 11: cost = 2, cell = nand2
14 node 12: cost = 1, cell = inv
15 node 13: cost = 5, cell = nand2
16 node 14: cost = 12, cell = nand2
17 node 15: cost = 2, cell = nand2
18 node 16: cost = 3, cell = inv
19 node 17: cost = 15, cell = nand3
20

```

Appendix: output_improve.txt

txt

```

1 Minimum total cost = 12
2
3 [Original nodes]
4 node 0 (PI): PI, cost = 0
5 node 1 (PI): PI, cost = 0
6 node 2 (PI): PI, cost = 0
7 node 3 (PI): PI, cost = 0
8 node 4 (PI): PI, cost = 0
9 node 5 (PI): PI, cost = 0
10 node 6 (PI): PI, cost = 0
11 node 7 (PI): PI, cost = 0
12 node 8 (NAND): cost = 2, cell = nand2
13 node 9 (INV): cost = 1, cell = inv
14 node 10 (NAND): cost = 5, cell = nand2
15 node 11 (NAND): cost = 2, cell = nand2
16 node 12 (INV): cost = 1, cell = inv
17 node 13 (NAND): cost = 5, cell = nand2
18 node 14 (NAND): cost = 10, cell = nand2
19 node 15 (NAND): cost = 2, cell = nand2
20 node 16 (INV): cost = 3, cell = inv
21 node 17 (NAND): cost = 12, cell = oai21 [OUTPUT]
22
23 [Inserted inverter-pair nodes]
24 node 18 (INV): cost = 1, cell = inv
25 node 19 (INV): cost = 0, cell = buf
26 node 20 (INV): cost = 1, cell = inv
27 node 21 (INV): cost = 0, cell = buf
28 node 22 (INV): cost = 1, cell = inv
29 node 23 (INV): cost = 0, cell = buf
30 node 24 (INV): cost = 3, cell = inv
31 node 25 (INV): cost = 2, cell = buf
32 node 26 (INV): cost = 0, cell = buf
33 node 27 (INV): cost = 1, cell = inv
34 node 28 (INV): cost = 1, cell = inv
35 node 29 (INV): cost = 0, cell = buf
36 node 30 (INV): cost = 1, cell = inv
37 node 31 (INV): cost = 0, cell = buf
38 node 32 (INV): cost = 1, cell = inv
39 node 33 (INV): cost = 0, cell = buf
40 node 34 (INV): cost = 3, cell = inv
41 node 35 (INV): cost = 2, cell = buf
42 node 36 (INV): cost = 0, cell = buf
43 node 37 (INV): cost = 1, cell = inv
44 node 38 (INV): cost = 3, cell = aoi21
45 node 39 (INV): cost = 4, cell = inv
46 node 40 (INV): cost = 3, cell = aoi21
47 node 41 (INV): cost = 4, cell = inv
48 node 42 (INV): cost = 1, cell = inv
49 node 43 (INV): cost = 0, cell = buf
50 node 44 (INV): cost = 1, cell = inv
51 node 45 (INV): cost = 0, cell = buf
52 node 46 (INV): cost = 3, cell = inv
53 node 47 (INV): cost = 2, cell = buf
54 node 48 (INV): cost = 9, cell = aoi21
55 node 49 (INV): cost = 10, cell = inv
56 node 50 (INV): cost = 2, cell = buf
57 node 51 (INV): cost = 3, cell = inv
58

```